Hall Ticket No: | | | | | | | | | | |

Course Code: A8703

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CSE

VARDHAMAN
COLLEGE OF ENGINEERING

III B.Tech I Semester Continuous Assessment - II, October - 2025

(Regulations: VCE-R22)

Machine Learning
(CSE & IT)

Date: 23rd October 2025 AN **Time: 2 Hours** **Maximum Marks: 30**

Answer all Questions in Part-A

Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	*CO Statement*	*Bloom's Level (L#)*
CO1	Identify the various concepts and challenges in machine Learning	L2
CO2	Select modelling and evaluation technique for handling real time data	L3
CO3	Use supervised and unsupervised learning algorithms for a given problem	L3
CO4	Examine supervised and unsupervised learning algorithms for analyzing data.	L4
CO5	Identify the various concepts of neural network to develop AI based applications	L2

Questions:

PART-A					
			CO#	*BL#*	*Marks*
1.	a)	What is likelihood probability? Give an example	CO3	L3	*1M*
	b)	Discuss any two suitable applications of Naïve Bayes Classifier.	CO3	L2	*1M*
	c)	Why the kNN algorithm is called a lazy learner?	CO3	L2	*1M*
	d)	Discuss Cluster Analysis and explain its purpose in grouping similar data points with a suitable example.	CO4	L3	*1M*
	e)	Explain how partitioning methods work in clustering.	CO4	L2	*1M*
	f)	What is the objective function of K-Means Algorithm?	CO4	L2	*1M*
	g)	What is the function of Synaptic weights in Artificial Neuron?	CO5	L2	*1M*
	h)	Write about ADALINE network model.	CO5	L2	*1M*
	i)	Write about the Tanh (Hyperbolic Tangent) activation function.	CO5	L2	*1M*
	j)	Discuss any 2 Weaknesses of K-Means	CO4	L2	*1M*

PART-B				
2.	Apply Naïve Bayes Algorithm, you need to classify the conditions when this team wins and then predict the probability of this team winning a particular match when Weather Conditions = Overcast, they won one of the last three matches, Humidity = Normal and they won the toss in the particular match.	CO3	L3	5M

Weather Condition	Wins in last 3 matches	Humidity	Win toss	Won match?
Rainy	3 wins	High	FALSE	No
Rainy	3 wins	High	TRUE	No
OverCast	3 wins	High	FALSE	Yes
Sunny	2 wins	High	FALSE	Yes
Sunny	1 win	Normal	FALSE	Yes
Sunny	1 win	Normal	TRUE	No
OverCast	1 win	Normal	TRUE	Yes
Rainy	2 wins	High	FALSE	No
Rainy	1 win	Normal	FALSE	Yes
Sunny	2 wins	Normal	FALSE	Yes
Rainy	2 wins	Normal	TRUE	Yes
OverCast	2 wins	High	TRUE	Yes
OverCast	3 wins	Normal	FALSE	Yes
Sunny	2 wins	High	TRUE	No

3.	What is the Random Forest algorithm? Explain how it overcomes the limitations of a Decision Tree using the concepts of bagging and ensemble learning. Include a diagram to support your explanation.	CO3	L2	5M
4.	Write the K-Medoids algorithm and illustrate its effectiveness in handling outliers with a suitable example.	CO4	L3	5M
5.	A fitness trainer wants to categorize participants based on their weekly workout duration (in minutes) to identify beginners, intermediates, and advanced trainees. The recorded durations are: [60, 75, 90, 105, 120, 135, 150, 165, 180, 195]. Apply the K-Means algorithm to form 3 clusters using the initial centroid values 75, 120, and 165.	CO4	L3	5M
6.	Illustrate with examples how non-linear activation functions enable neural networks to learn complex patterns.	CO5	L3	5M
7.	Draw the architecture of a Feed Forward Neural Network and discuss the roles of the input layer, hidden layers, and output layer in learning and prediction.	CO5	L2	5M

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SCHEMA OF VALUATION

Name of the Subject	: Machine Learning	Subject Code	: A8703
Programme	: B.Tech	Year	: III rd yr 2025-26
Examination	: Regular / Supplementary	Date of Examination	: 23/10/25
Max. Marks	: 30 marks	Time of Examination	: AN

1 a) Likelihood Probability – Probability of predictor B given class A (1m)

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

P(B/A) is likelihood Probability.

b) Any two Naive Bayes Classifier Applications (1m)
1) Email filtering
2) Network Security

c) KNN called as lazy learner – (1m)
- KNN does not learn during training phase but uses training data during testing phase to classify test samples.

d) Cluster Analysis with Example (1m)
- Finding groups of objects such that objects in a group will be similar to one another and different from the objects in other groups

Ex: Customers segmentation

e) Partitioning Method (1m)

- Uses mean or mediod to represent cluster centre
- Adopts distance based approach to refine clusters.

f) K-Means objective function (1m)

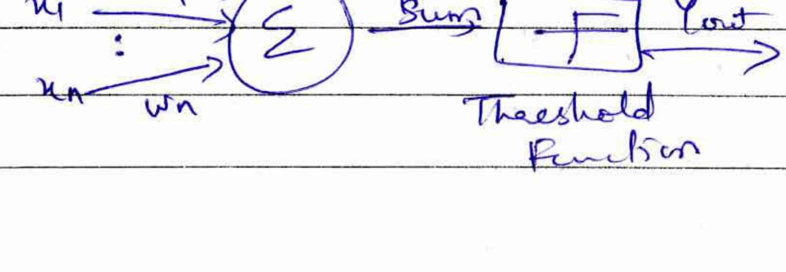
- To maximize the homogeneity within the clusters & to maximize the difference between the clusters.

g) Synaptic Weights in Artificial Neuron (1m)

- Synapse (weights): point of connections to another.
- A signal x_i forms the i/p to the i^{th} synapse having weight w_i.
- Value of w_i may be positive or negative

(h) ADALINE N/w Model (1m)

It is an early single layer ANN developed by professor Bernard Widrow of Stanford University.



(8) Tanh Activation Function (1m)

- Hyperbolic tangent function is continuous activation function, which is bipolar in nature.

(9) Any two weaknesses of K-means (1m)

- K is selected randomly, thus may not find optimal set of clusters
- For selecting initial clusters requires some experience of the user.

2. Problem ~~steps~~ solution steps (5m)

Weather = overcast, Wins in last 3 matches = 1 win

Humidity = Normal, win toss = TRUE

Step1: calculate prior probabilities

Won = 8

Lost = 6

Total = 14

$P(\text{win}) = 8/14 = 0.571$

$P(\text{Lose}) = 6/14 = 0.429$

Step2: Calculate ~~likelihood~~ conditional probabilities

For win = Yes (8 records)

Weather = overcast = 3/8 = 0.375

wins = 1 win 2/8 = 0.25

Humidity = Normal 5/8 = 0.625

Toss = TRUE 3/8 = 0.375

$P(X/\text{win}) = 0.375 \times 0.25 \times 0.625 \times 0.375$

$= 0.02197$

For win = NO (6 records

Weather = overcast . = 0 no·1

wins = 1 win = 1/6 = 0.167

Humidity = Normal 2/6 = 0.33

TOSS = TRUE 3/6 = 0.5

P(X/Lose) = 0.1 × 0.167 × 0.33 × 0.5 = 0.00278

Step3: Posterior probabilities

P(win/X) ∝ P(X/win) P(win) = 0.02197 × 0.571

= 0.01255

P(Lose/X) ∝ P(X/Lose) P(Lose) = 0.00278 × 0.429

= 0.00119

P(win/X) > P(Lose/X)

Final Prediction : WIN. with 91% probability

$\because P(win/X) = \frac{0.01255}{0.01255 + 0.00119} = 0.913$

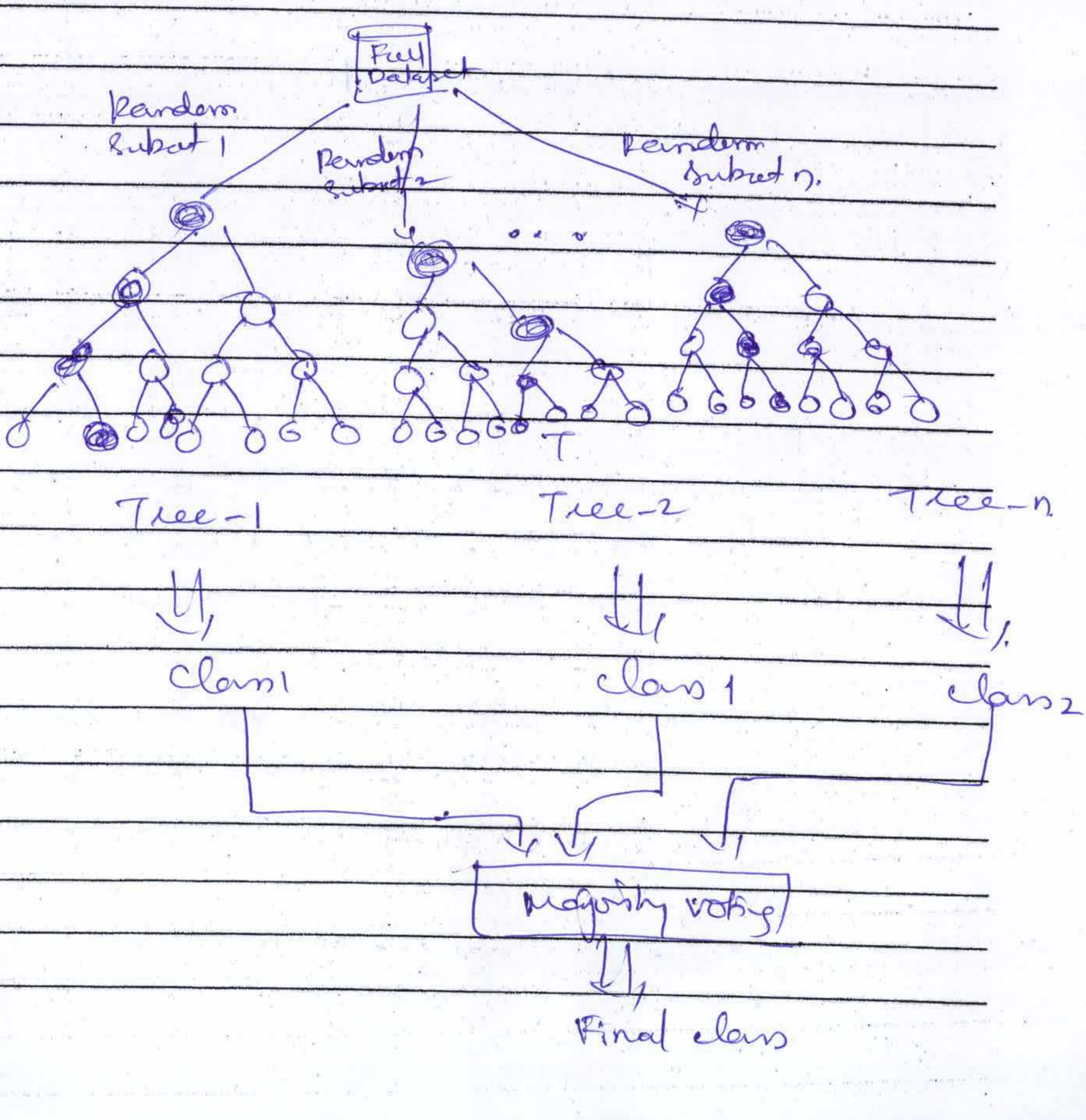
(3) Random Forest Algorithm – (3m)

Diagram – (1m)

DT Limitation (1m)

Random Forest (RF)

- Random Forest is an ensemble classifier, that combines many decision tree classifiers.
- Bagging is applied with different feature sets
- Train the trees such that contribution from each feature comes in a no. of models.
- After the RF is generated by combining trees (majority vote) usually which is better than individual decision tree models.



Ans. Strengths of RF

- It runs efficiently on large & expansive data sets
- It has robust method for estimating missing data and maintains precision when a large proportion of data is ~~present~~ absent.
- It has powerful techniques for balancing errors in class population of unbalanced data sets

(e) K-medoid Algorithm - 4m
Handling outliers - 1m

Algorithm steps

1) Arbitrarily choose K objects in D (no. of clusters) as initial representative objects.
2) Repeat
3) Assign each remaining object to the cluster with the nearest representative object.
4) Randomly select non representative object; Orandom.
5) Compute the total cost S, of swapping representative object Oj with Orandom.
6) If S < 0 then swap Oj with Orandom to form the new set of k representative objects.
7) until no changes.

Effectiveness on Handling Outliers

Dataset points = {2, 3, 4, 5, 20}

K=2

Using K-Medoids.

- Medoids are actual points
- Suppose medoids = {3, 20}
- Outlier (20) becomes its own cluster, while {2, 3, 4, 5} stay in one compact cluster.
- The medoid (3) remains unaffected by extreme value '20'

(5) Problem solution steps (5m)

Step1: Given data.

60, 75, 90, 105, 120, 135, 150, 165, 180, 195

Initial centroids:

$C_1 = 75$ $C_2 = 120$ $C_3 = 165$

Step2: Compute Distance & Assign clusters

Data points	C1	C2	C3	Assigned Cluster
60	15	60	105	C1
75	0	45	90	C1
90	15	30	75	C1
105	30	15	60	C2
120				
135				
150				
165				
180				
195				

Clusters: after Iteration 1

C1 : {60, 75, 90}

C2 : {105, 120, 135}

C3 : {150, 165, 180, 195}

Step 2 – compute new centroids

$$C_1' = \frac{60+75+90}{3} = 75$$

$$C_2' = \frac{105+120+135}{3} = 120$$

New centroids : 75, 120, 172.5

Step 3 : Reassign using centroids 75, 120, 172.5

60	15	60	112.5	C1
75	0	45	97.5	C1
90	15	30	82.5	C1

clusters remain the same.

Final clusters

cluster 1 : 75

60, 75, 90

cluster 2 : 120

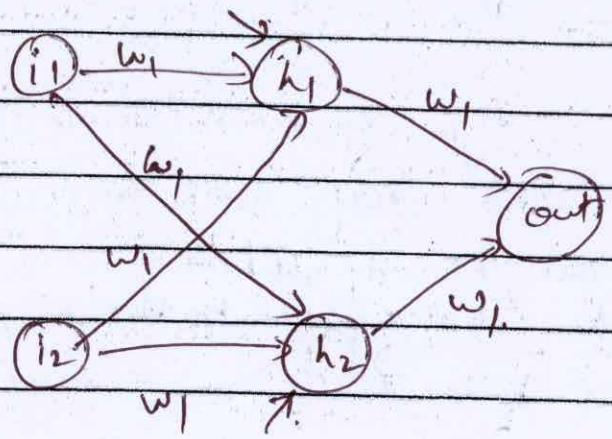
105, 120, 135

cluster 3 – 172.5

150, 165, 180, 195

(b) Non-linear Activation Function - (5m)

- Non-linear activation functions enable neural n/ws to learn complex patterns by introducing non-linearity, which allows the n/w to create complex, curved decision boundaries instead of just straight lines



The hidden layer o/ps are:

$h_1 = i_1 \cdot w_1 + i_2 \cdot w_3 + b_1$

$h_2 = i_1 \cdot w_2 + i_2 \cdot w_4 + b_2$

The output before activation is:

output: $h_1 \cdot w_5 + h_2 \cdot w_6 + bias$

without activation, these are linear equations.

To introduce non-linearity we can apply activation function such as sigmoid

$$\sigma(x) = \frac{1}{1+e^{-z}}$$

Final output $= \sigma(h_1 \cdot w_5 + h_2 \cdot w_6 + bias)$

(7) Feed Forward Neural N/w (FNN) —(3m)
Diagram —(2m)

The multi-layer feed forward n/w is quite similar to the single-layer feed forward n/w, except for the fact that there are one or more intermediate layers of neurons between the i/p and o/p layers.

- FNN is a type of ANN in which information flows one way i.e. from i/p to output.
- No loops or feedback.
- Used for tasks like pattern recognition (eg: image & speech).